

Alexis (Lexi) Arlen

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EDUCATION

Stanford University

Ph.D. in Earth System Science

Advisor: Prof. Earle Wilson

Dissertation: *Fracture and Floes: Emergent Sea Ice Behavior from Observations and Discrete Element Modeling*

Stanford, CA

expected June 2027

Washington University in St. Louis

B.A. in Mathematics; B.A. in Geophysics

St. Louis, MO

May 2022

PUBLICATIONS

Wilson, E. A., **Arlen, L.**, Campbell, E. C. (2026). Recent extremes in Antarctic sea ice extent modulated by ocean heat ventilation. *Proceedings of the National Academy of Sciences*, 123(14), e2530832123.

Nabelek, P., **Arlen, A.**, Fromcke, T. (2024). On periodic and finite genus solutions to the integrable Kaup–Broer system for capillary waves. *Applied Numerical Mathematics*, 199, 123–135.

PUBLICATIONS IN PROGRESS

Arlen, L., Wilson, E. A., Dansereau, V., Lai, C. Y., Meng, Y. A simple bonded discrete element model for sea ice breakup. *EGUsphere* [preprint], in review at *The Cryosphere*.

Arlen, L., Wilson, E. A., Buckley, E. M., Watkins, D. M., Sea ice temperature determines the sizes of the biggest floes. *In prep.*

HONORS AND AWARDS

Best Student Poster, AMS 18th Polar Meteorology and Oceanography Conference 2025

McGee–Levorson Grant, Stanford University 2023 | \$2,500

NSF Graduate Research Fellowship 2022 | \$159,000

Travel Award, Gulf Coast Undergraduate Research Symposium 2021

PRESENTATIONS

“What sets the sizes of the biggest sea ice floes?” Talk, *CalGFD*, Los Angeles, CA, September 2026.

“A surprisingly capable minimal bonded discrete element model for sea ice.” Talk, *EGU General Assembly*, Vienna, Austria, May 2026.

“The Sensitivity of the Emergent Sea Ice Floe Size Distribution to Wind Stress and Ice Strength.” Poster, *Scale-Aware Sea Ice Project (SASIP) Annual Meeting*, Providence, RI, June 2025.

“The Sensitivity of the Emergent Sea Ice Floe Size Distribution to Wind Stress and Ice Strength.” Poster, *AMS 18th Conference on Polar Meteorology and Oceanography*, Denver, CO, May 2025.

“Observations of Ocean Surface Waves in the Chukchi Sea Marginal Ice Zone.” Poster, *Ocean Sciences Meeting*, New Orleans, LA, February 2024.

“In Situ Observations of Summertime Surface Wave Propagation in the Chukchi Sea Marginal Ice Zone.” Poster, *AGU Fall Meeting*, San Francisco, CA, December 2023.

“Observations of Ocean Surface Wave Propagation in the Chukchi Sea Marginal Ice Zone.” Poster, *Graduate Climate Conference*, MIT/WHOI, Woods Hole, MA, November 2023.

“One-Gap Solutions to the Kaup–Broer System.” Talk, *Gulf Coast Undergraduate Research Symposium*, Rice University, Houston, TX, October 2021.

FIELDWORK

R/V Sikuliaq — Tale of Three Systems (ToTS) Cruise

Chukchi Sea

PI: Dr. Kevin Arrigo

June–July 2023

Designed and executed experiments to study surface wave attenuation in sea ice and seasonal changes to the sea ice floe size distribution with UAV surveys. Supported sediment trap and multi-corer operations. 45 days.

Cascadia Subduction Zone Seismic Imaging Campaign

Coastal Oregon

PIs: Dr. Anne Tréhu, Dr. Emilie Hooft, Dr. Kevin Ward, Dr. Erin Wirth

May–July 2021

Supported the deployment and recovery of 750 nodal seismometers across Oregon’s Coast Range to image the structure of the Cascadia Subduction Zone. Launched and maintained a public field blog documenting the campaign.

ACADEMIC AND UNIVERSITY SERVICE

Scientific Community Service

AGU Fall Meeting

2026

Session co-convenor, “Ice-Ocean Interactions in High-Latitude Environments”

University Service

Organizer for:

- Stanford Ice quarterly research seminar and social, 2024–present
- California Geophysical Fluid Dynamics (CalGFD) Meeting, 2024
- FLOES virtual monthly sea ice research seminar, 2024

Panelist for:

- Earth System Science Qualls Panel for early graduate students, 2024 & 2025)
- Oceans Department Qualls Panel for early graduate students, 2024

TEACHING

Climate Change: Projections and Uncertainty

Summer 2025 & 2026

Original two-week course for high school students, Stanford Summer Pre-Collegiate Institute

Polar Climate Dynamics (ESS 248)

Spring 2024

Teaching assistant and guest lecturer, Prof. Earle Wilson, Stanford University

Calculus II (Math 132)	Spring 2022
Teaching assistant and discussion section leader, Prof. Silas Johnson, Washington University	
Natural Disasters (EPSc 131)	Spring 2021
Teaching assistant, Profs. Mike Krawczynski and Doug Wiens, Washington University	
Calculus I (Math 131)	Fall 2020 & 2021
Teaching assistant and discussion section leader, Prof. Silas Johnson, Washington University	

MENTORING

Ella Walsh	Summer 2025
<i>Stanford Sustainability, Engineering and Science Undergraduate Research Program</i>	
Influence of floe–floe collisions and fracture on sea ice velocity fluctuations in a discrete element model	
Allie Skalnik	Summer 2024
<i>Stanford Sustainability, Engineering and Science Undergraduate Research Program</i>	
Antarctic sea ice variability from reanalysis, passive microwave remote sensing, and idealized 1-D modeling	
Chloe Cheng	Summer 2023
<i>Stanford Sustainability, Engineering and Science Undergraduate Research Program</i>	
Segmentation of SAR imagery to assess storm influence on the sea ice floe size distribution	
Science Small Groups Mentor	Fall 2024
Weekly science mentoring for community college students	
Stanford-Summit Tahoma Expeditions Program Mentor	January 2025
Hosted a visiting high school student in the Polar Ocean Dynamics lab for two weeks	
ESS Peer Mentor	Fall 2023
Mentor for incoming graduate students, Stanford	
Matched Academic Mentor for Calculus Courses	2019–2022
Weekly small-group mentoring for students from disadvantaged backgrounds, Center for Teaching and Learning, Washington University	

OUTREACH

- Lead instructor for weekly Earth science lessons with Sprout Up, Escondido Elementary School, Fall 2025
- Instructor for biweekly geology field trip with GeoKids, Stanford University, Fall 2024
- Science fair judge, Escondido Elementary School, April 2026
- Climate and sea ice talk, Palo Alto High School, January 2026
- Fluid dynamics science demo, The Gateway School, November 2025
- Career panelist, The Athenian School, March 2023 and April 2025